Abdullah Mohammad Aref

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Professional Summary

Software engineer and BSc Computer Science student with strong foundations in TypeScript, Python, and full-stack web architectures. Experienced in building responsive web applications, designing RESTful APIs, and developing automated data pipelines. Applies strong component architecture and asynchronous data flows to build modular developer tools and platform services. Committed to software reliability, clean code standards, and strict data integrity.

Professional Experience

Freelance Software Engineer | Berlin, Germany 09/2025 – 09/2026

- Rebuilt legacy Microsoft Access spreadsheets into a modern Django and React platform with a normalized 26-model relational schema, centralizing 2,000+ vehicle records and automating bank transaction ingestion.
- Integrated the mobile.de REST API for automated vehicle listing and engineered a PDF engine generating 6 legally compliant German sales documents in seconds.
- Engineered secure REST APIs with role-based permissions, containerized services using Docker, and deployed on cloud infrastructure with automated CI/CD pipelines to ensure reliable production uptime.

Education

Gisma University of Applied Sciences | BSc Computer Science, Berlin, Germany 2023 – 2027

Projects

Flow (AST & Codebase Mapping Engine) | Python, TypeScript, React, Scikit-learn 04/2026 – 06/2026

- Built a static analysis engine in Python parsing Abstract Syntax Trees (AST) across 1,500+ source files to convert unstructured code into structured graph datasets.
- Engineered an end-to-end data pipeline that balances skewed class distributions and classifies records with a 76% macro F1-score.
- Developed an interactive React and TypeScript interface with dynamic search and metadata filtering to trace relational entity dependencies in real time.

E-Commerce Web Application | Vue.js, JavaScript, Node.js, HTML5, CSS3 01/2026 – 03/2026

- Developed a full-stack e-commerce store with Vue.js and Node.js, designing a responsive UI across all devices.
- Built REST APIs to support CRUD operations and implemented secure user authentication.
- Integrated PayPal Sandbox API to process payments, managed user and cart state with Pinia, and optimized image processing with Multer and Sharp.

Cars Resale Value Prediction | Python, Scikit-learn, Pandas, NumPy 07/2026 – 09/2026

- Engineered an end-to-end data processing pipeline in Python on 37,800+ automobile records to model market valuation trends and depreciation drivers.
- Cleaned and normalized messy tabular datasets, imputed missing attributes, encoded categorical variables, and removed collinear predictors using correlation matrix filtering.
- Trained and evaluated Scikit-learn regression models via cross-validation, achieving an R^2 score of 82.8% in predicting used-vehicle market resale values.

Skills & Competencies

Programming Languages: TypeScript, JavaScript, Python, SQL, Node.js, Java, R.
Frameworks & Libraries: Next.js, React, Node.js, Express.js, Django, Scikit-Learn, Pandas.
Databases & Cloud: PostgreSQL, Supabase, Docker, Git, GitHub Actions, Vercel, AWS, SQLite3.
Core Concepts: Full-Stack Architecture, APIs, API Design, Real-Time Data Flows, Rapid Prototyping, Automated Testing, AI Agent Workflows.
Soft Skills: High Agency, Proactive Problem Solving, Clear Technical Communication, Quick Grasp.

Certificates & Awards

2nd Place Winner | Nexus Hackathon: The Odyssey (Gisma University) 06/2026

- Collaborated in a 3-person team to rapidly brainstorm an innovative technical solution, build a working prototype with AI development tools in three hours, and pitch the live prototype to judges.

Languages

English C1 | German A2